

# ASSIGNMENT-2 DATA & METHODOLOGY

## Brain Tumor Detection System

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### I. RESEARCH QUESTION & SCOPE OF THE PROJECT

#### A. Research Question

Is it possible to classify Brain MRI scans clearly stating as tumor or non-tumor with high degree of accuracy using CNN (Convolutional Neural Networks) based system?

#### B. Why it Matters

As we all know, Brain Tumors are serious medical emergencies, if the brain tumors gets detected earlier they can eventually be cured & improve the treatments. MRI scans are being installed manually by radiologists in a normal hospital and this manual installation takes time & in worst case scenarios the installation may have some inconsistencies due to heavy workload.

An AI-Based Brain tumor detection system does not mean to replace doctors. But, it can work as a secondary assistant that can assist the doctors by giving fast & at the same time consistent predictions based on MRI scans.

#### C. Definition of The Task

This problem can be classified as Binary Classification Problem (Binary Classifier) which will state: If Class 0 then No Tumor, If Class 1 then Tumor.

The model predicts a probability for each inserted image ( $x$ ), the formula used is:

$$\hat{p}(y = 1|x) \in [0, 1]$$

A decision threshold which we called it as ( $\tau$ ) is used using the formula:

$$\hat{y} = \begin{cases} 1, & \hat{p} \geq \tau \\ 0, & \hat{p} < \tau \end{cases}$$

### II. SELECTING THE DATASETS, RELATIVITY & THE QUALITY OF DATA

#### A. 2.1 Why I have selected this dataset

In this project I'm using labelled MRI scan images as tumor/no-tumor from public datasets. This choice is perfect for the following three reasons:

- 1) **Technical Relativity:** CNNs (Convolutional Neural Networks) are best & are very effective for the image-based classification tasks. Especially when the patterns like image textures & edges matters. [1], [3]
- 2) **Clinical Relativity:** MRI scan images are widely used in clinics & hospitals and are reliable for Brain-Tumor-Detection or Brain-Tumor-Screening.

- 3) **Transparent Relativity:** Considering public datasets, allows us to set a benchmark or we can say a target for this work against the previous work. [1]–[5]

My, choices of dataset reflect the previous studies that showed positive results using and at the same time strong performance by using both the approaches that were CNN based as well as the learning based on MRI image Classification. [1], [3]

#### B. 2.2 Dataset Challenges & Data Quality Checks

Challenges involved in datasets can be:

- 1) Brightness differentiating between the MRI images.
- 2) Different image sizes.
- 3) Type of scanner used.
- 4) Depth or intensity of black background.

This, type of challenges are usually involved in medical imaging diagnosis.

Before training following Data Quality Checks are performed:

- 1) Detection of corrupted files.
- 2) Detection of duplicated images.
- 3) Balancing class analysis.
- 4) Equally distribution of pixels.
- 5) Fix size for MRI images.

The above data quality checks are important because the Convolutional Neural Networks (CNN) can also include irrelevant background patterns or borders. So, this data quality checks reduces the risk and improves the accuracy.

After overcoming the challenges and performing the data quality checks the system workflow we get is: Input: 2D Brain MRI Images. Output Labeled As: Tumor or No Tumor.

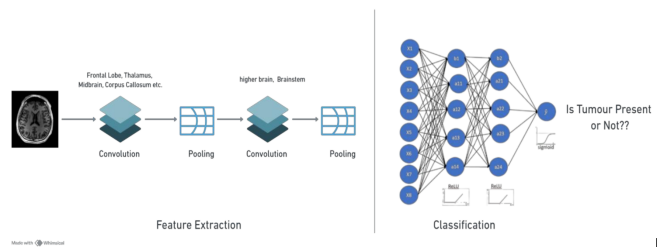


Fig. 1. System workflow from feature extraction to tumor classification.

### C. 2.3 Data Visualization

**Figure 2: Tumor Distribution Chart:**

The chart basically shows the number of tumors vs no tumors sample. As we can see from my chart, the images having tumor are higher as compared to non-tumor which shows a bit imbalance. As a result this imbalance will help to plan the accuracy of the model during evaluation that is MCC, F1, Recall, etc.

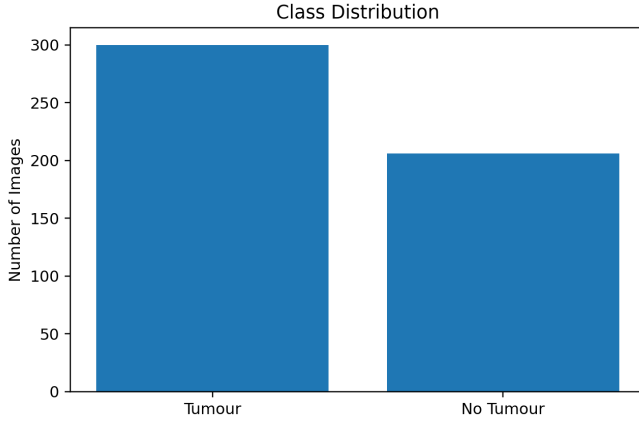


Fig. 2. Tumor distribution chart (Tumor vs No Tumor).

**Figure 3: Histogram showing Image Resolution:**

This figure basically shows the height and width of the images which are provided as input. This basically tells us image resizing should be done before training.

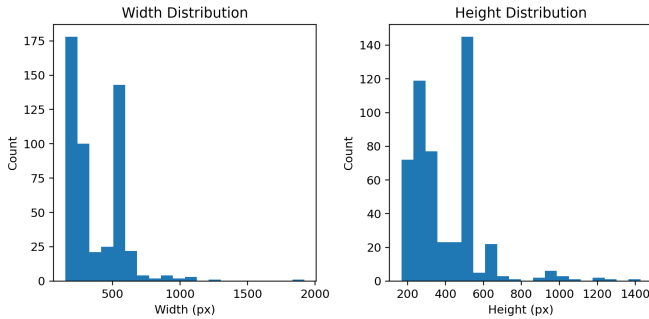


Fig. 3. Histogram showing image resolution (width and height distribution).

**Figure 4: Before Vs After Pre-Processing of MRI Images:**

This figure basically shows the original raw images before but after it shows region-focused and cropped to only the brain part ignoring the background. Which shows that Pre-Processing keeps the affected brain content and removes the unnecessary background.

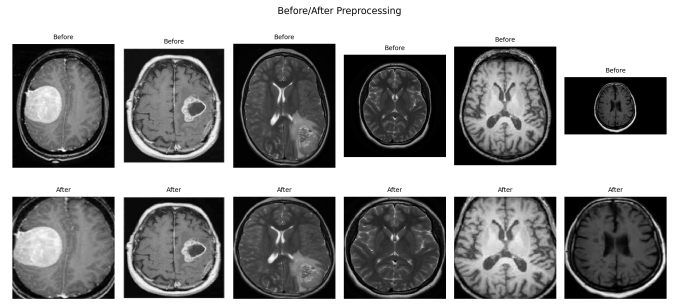


Fig. 4. Before vs after pre-processing of MRI images.

### III. DATA PREPARATION & PREPROCESSING PIPELINE

There are certain steps involved for the preprocessing pipeline which are designed to match the inputs based on CNN requirements.

#### A. 3.1 Steps Involved

- 1) It starts by reading the MRI image using OpenCV.
- 2) It converts the image in the required format like grayscale workflow.
- 3) Then, it crops the image and majorly focuses on the brain affected area.
- 4) The MRI image is resized to fixed input size that is  $50 \times 50$ .
- 5) Then, it equalizes the pixels based on pixel values.
- 6) Finally, the labels are encoded into binary form.

#### B. 3.2 Data Splitting Strategy

The Dataset is divided into three phases: Training Set Validation Set Testing Set. This shows the following workflow:

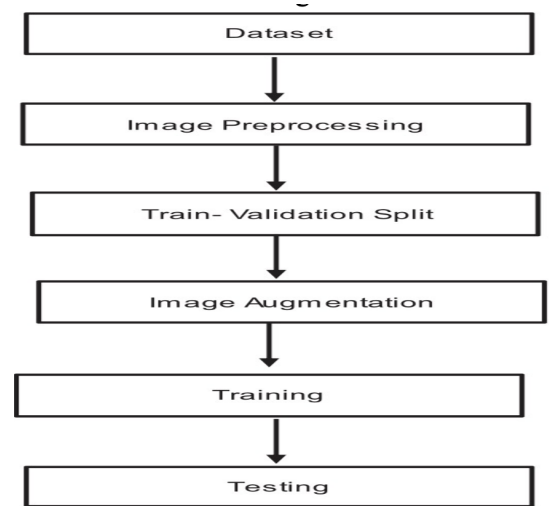


Fig. 5. Data preprocessing, split, augmentation, training, and testing workflow.

### IV. DEEP LEARNING METHODOLOGY

#### A. 4.1 Model Chosen: Convolutional Neural Network (CNN)

For Brain-Tumor-Detection System I am using CNN model because MRI Images is our input data & CNNs have the

specialty to adapt patterns like the background, shapes and edges properly.

CNN basically learns in different stages like:

- 1) **Convolutional layers:** These layers are nothing but minute scanners which moves over the images and detects patterns like dark/bright contrasting, edges and unnecessary regions (noise).
- 2) **ReLU Signaling:** Once the convolutional steps is performed,  $\text{ReLU}(\max(0, x))$  inherits the positive signals and removes the unnecessary not required weak signals (weaker negatives).
- 3) **Pooling:** Pooling basically reduces the size of the image so that, calculations become smaller, model targets the strongest features and also at the same time reduces the overfitting risks.
- 4) **Layer Learns:** Early layers basically learn simple features like edges but at the same time complex layer (deeper layers) learns complex medical structures.
- 5) **Connecting Layers:** The outputs of these layers are combined and gives the final decision.
- 6) **Sigmoid Layer (Output Layer):** The Sigmoid layer is basically the final probability output between 0 & 1: If it is close to 1 then likely a Tumor. If it is close to 0 then likely no Tumor.

#### B. 4.2 Why I have chosen CNN instead of RNN/Transformer

CNN basically gives us a balanced strong approach of simplicity (easier to explain train, test & debug), Efficiency (lower computation power is required as compared to RNN), & Deployability (integrating with Streamlit app makes it easier). [1], [3]

**Why not RNN:** RNNs are basically built for sequential data like speech or text. Based on my project it basically uses MRI images which are static and not ordered in a particular sequence. So, RNN is not a perfect fit for this project.

**Why not Transformer like ViT:** Transformers like Vision Transformers are considered to be as very powerful and required larger datasets, more computational power, more GPU memory, longer training & testing time & lastly for integrating the UI it can be a tougher job and may require other UI platform other than Streamlit. According to me, for this university deep learning project this will take a lot of amount of time and can increase the complexity.

#### C. 4.3 Training Pipeline Used

The training pipeline basically starts with a formula which involves Binary Cross Entropy (Loss), Optimizer, L2 Regularization, Validation Control (For Early Stopping & Saving The Best Checkpoint) and lastly for Executing Mini Batch Training. The formula used is:

$$L_{BCE} = -\frac{1}{N} \sum_i [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)]$$

#### V. LINKS TO EXISTING RESEARCH

My project Brain Tumor Detection System perfectly aligns with the previous works and is advanced by integrating Streamlit UI:

- 1) VCG based approaches and other transfer learning methods are considered as stronger alternative but with limited data. [3]
- 2) CNN-based classification of MRI images has significantly shown higher performance. [1]
- 3) The Hybrid CNN with SVM approach also shows good amount of generalization power in some of the settings. [5]
- 4) Capsule Network & U-Net studies have shown that this segmentation & spatial modelling can be introduce towards future scope. [2], [4]

#### VI. EVALUATION METRICS WITH FORMULAS

The TP (True Positive), TN (True Negative), FP (False Positive), and FN (False Negative) be the confusion matrix counts. So, the formulas which I'm using for calculating Precision, Recall, Accuracy, Specificity, F1, MCC, & Brier are:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$\text{Brier} = \frac{1}{N} \sum_i (\hat{p}_i - y_i)^2$$

#### VII. WHY I HAVE SELECTED THIS METRICS

- 1) Precision + Specificity: Using both together controls the False Positive.
- 2) F1 + MCC: This are useful for controlling the imbalancing between the class.
- 3) Recall: This can be considered as important formula for this project because this formula helps to reduce the tumor cases which are missed.
- 4) AUC/Loss/Brier: This is basically used to show the overall quality of the model which shows that in this I have not been depended only on one threshold.

#### REFERENCES

- [1] Deepa, A., Shanthini, M., et al. (2020). Brain Tumor Detection and Classification Using CNN.
- [2] Isensee, F., Jaeger, P. F., et al. (2019). Brain Tumor Segmentation Using U-Net Based Deep Learning (BraTS).
- [3] Swati, Z. N. K., et al. (2019). Classification of Brain Tumors Using Transfer Learning Techniques.
- [4] Afshar, P., Mohammadi, A., Plataniotis, K. N. (2020). Deep Learning for Brain Tumor Classification and Segmentation (Capsule Networks).
- [5] Paul, J. S., Chatterjee, J. (2021). Automated Brain Tumor Detection Using Hybrid CNN-SVM Approach.